

Israel News & Political Intelligence

Comprehensive Backend API Documentation & Logic Manual

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Framework: FastAPI (Python) & MongoDB & Redis Cache

Author: Antigravity AI Assistant & Developer Team

This document outlines the business logic, request schema configurations, and sample responses for all routing layers.

System Architecture & Implementation Overview

The Israel News & Political Intelligence API is a highly modular, real-time data aggregation and analytical engine. Built on **FastAPI**, it handles high-throughput scraping, concurrent RSS parsing, AI bias evaluations, Knesset intelligence syncs, and Q&A; interactions.

Key Infrastructure Pillars:

- **FastAPI & Pydantic:** Strictly validates request schemas and enforces response typing for client guarantees.
- **MongoDB Storage:** Persists crawled news categories, user crowdsourced votes, MP profiles, and Knesset voting tally indexes.
- **Caching Layers:** Key endpoints implement dynamic TTL cache locks (news, Knesset bills) in Redis/Memory to prevent rate limits.
- **Multi-Model AI Integration:** Implements a tiered analysis structure. Free tiers fall back to token/regex rule baselines; Pro uses GPT-4o-mini; Platinum activates an ensemble of Perplexity, Gemini, and OpenAI models with weighted average consensus.

1. Health & Diagnostics

GET / — Root Health & Metadata

Backend Logic: Provides a simple check on root URL to verify FastAPI server is running. Returns service status, version, and the interactive FastAPI documentation path.

Sample Input / Params: None

Sample Output / Response:

```
{
  "status": "ok",
  "version": "3.0.0",
  "docs": "/docs"
}
```

GET /health — System Liveness & Health

Backend Logic: Returns the system status alongside a UTC timestamp. Useful for Kubernetes liveness probes or automated uptime trackers.

Sample Input / Params: None

Sample Output / Response:

```
{
  "status": "healthy",
  "timestamp": "2026-06-06T03:16:33.456Z"
}
```

GET /diagnostic/status — External Services Status Check

Backend Logic: Inspects settings configuration environment variables for OpenAI, Gemini, Claude, and Perplexity keys. Reports whether each service API is configured or not configured.

Sample Input / Params: None

Sample Output / Response:

```
{
  "status": "ok",
  "services": {
    "openai": "configured",
    "gemini": "not configured",
    "claude": "not configured",
    "perplexity": "configured",
    "mongodb": "check via /political/mps or similar endpoint"
  }
}
```

GET /diagnostic/openai/test — OpenAI Connectivity Check

Backend Logic: Initializes an AsyncOpenAI client using the configured OPENAI_API_KEY. Sends a short test prompt 'Say ok' to model 'gpt-4o-mini' with max_tokens=5. Returns success metrics (model, tokens) or detailed troubleshooting steps on failure.

Sample Input / Params: None

Sample Output / Response:

```
{
  "status": "ok",
  "message": "OpenAI API is reachable and key is valid",
  "model": "gpt-4o-mini-2024-07-18",
  "tokens_used": 5
}
```

GET /diagnostic/gemini/test — Gemini Connectivity Check

Backend Logic: Configures google.generativeai with GEMINI_API_KEY, loads model 'gemini-pro', and triggers generate_content('Say ok'). Returns response text on success or instructions to generate an API key on failure.

Sample Input / Params: None

Sample Output / Response:

```
{
  "status": "ok",
  "message": "Gemini API is reachable and key is valid",
  "response": "ok"
}
```

2. News Aggregation

GET /categories — List News Categories

Backend Logic: Extracts and returns all active news categories supported by the application from the RSS_FEEDS map configuration keys.

Sample Input / Params: None

Sample Output / Response:

```
{
  "categories": [
    "social", "economy", "defence", "education", "community",
    "political", "positive", "sport", "culture", "environment",
    "science", "international", "knesset", "arabic"
  ],
  "count": 14
}
```

GET /sources — List News Sources

Backend Logic: Retrieves preconfigured news sources. If detailed=true, lists sources with meta-information (language, country, default bias, baseline credibility score). If detailed=false, yields basic arrays of source names grouped by language.

Sample Input / Params: ?language=english&country=Israel&detailed=true
(Parameters are optional. detailed defaults to false)

Sample Output / Response:

```
{
  "sources": [
    {
      "name": "Times of Israel",
      "url": "https://www.timesofisrael.com",
      "feed_url": "https://www.timesofisrael.com/feed/",
      "country": "Israel",
      "language": "English",
      "bias": "center",
      "credibility": 0.85,
      "category": "general"
    }
  ],
  "total": 1
}
```

GET /news/all — Fetch All News Categories

Backend Logic: Concurrently fetches RSS feeds for all categories using `asyncio.gather`. Cleans HTML tags, filters opinion pieces and blocked sources, removes duplicate links, and applies category filters (e.g. excluding negative news for positive categories). Optional tiered AI analyses are run concurrently if `with_analysis=true`.

Sample Input / Params: `?limit=10&user_tier=free&with_analysis=false`

Sample Output / Response:

```

{
  "economy": {
    "meta": {
      "title": "Israeli News - Economy",
      "description": "Top articles",
      "link": "https://talmicahel.com"
    },
    "total": 1,
    "articles": [
      {
        "title": "Bank of Israel maintains rate",
        "link": "https://www.globes.co.il/news/article.1001",
        "description": "The central bank decided to hold interest rates...",
        "pub_date": "2026-06-06T07:00:00Z",
        "source": "Globes",
        "source_url": "https://globes.co.il"
      }
    ]
  }
}

```

GET /news/knesset — Retrieve Knesset Bills & Fallbacks

Backend Logic: Tries to fetch Knesset bill legislation records from the official Knesset OData API. If unavailable, falls back to Google News RSS Kneset queries. If that also fails, queries Wikipedia search APIs for summaries. If all fails, returns an empty array. Caches the response.

Sample Input / Params: `?limit=20`

Sample Output / Response:

```

{
  "source": "knesset.gov.il OData API",
  "official_api_reachable": true,
  "official_api_notice": "Connected to official Knesset OData API",
  "total": 1,
  "bills": [
    {
      "id": 20456,
      "name": "State Budget Law 2026",
      "name_hebrew": "תקציב המדינה 2026",
      "status": "Approved Third Reading",
      "sub_type": "Government Bill",
      "last_updated": "2026-06-05T18:30:00",
      "initiator": "Ministry of Finance"
    }
  ]
}

```

GET /news/{category} — Fetch Specific Category News Feed

Backend Logic: Fetches RSS feeds for a given category (e.g. economy, defence). Parallel parses XML feeds, runs deduplication by URL, filters blocked sources and opinions. If with_analysis=true, initiates AI analysis matching the user tier (Free: Rule-based, Pro: GPT-4o-mini, Platinum: Ensemble). Uses MongoDB caching.

Sample Input / Params:

Path: /news/economy

Query Params: ?limit=20&exclude_negative=false&user_tier=pro&with_analysis=true

Sample Output / Response:

```
{
  "meta": {
    "title": "Israeli News - Economy",
    "description": "Top articles from preconfigured feeds",
    "link": "https://talmicahel.com"
  },
  "total": 1,
  "articles": [
    {
      "title": "Israel Tech Exports Surge",
      "link": "https://www.timesofisrael.com/tech-surge",
      "description": "Exports of high-tech services rose by 12%...",
      "pub_date": "2026-06-06T08:00:00Z",
      "source": "Times of Israel",
      "source_url": "https://timesofisrael.com",
      "guid": "art-tech-123",
      "sentiment": "positive",
      "bias": "center",
      "bias_score": 0.12,
      "bias_types": ["source leaning"],
      "bias_category": "Objective Reporting",
      "credibility_score": 0.85,
      "credibility_label": "verified",
      "fact_check_score": 0.77,
      "summary_hebrew": "■■■■■ ■■■■■■ ■■■■■■■■ ■■■■ ■-12 ■■■■■■.",
      "topics": ["economy", "technology"],
      "claims": ["High-tech services exports rose by 12%"],
      "factual_points": ["Central Bureau of Statistics released trade figures today"]
    }
  ]
}
```

3. AI Analysis & Voting

POST /ai/analyze — Analyze Article Bias & Sentiment

Backend Logic: Analyzes article bias, sentiment, and credibility. Tiered strategy applied: - free: Rule-based analysis checking keyword list and seed dictionary leanings. - pro: Calls GPT-4o-mini with structured JSON parsing prompts. Falls back to rule-based on failure. - platinum: Queries GPT-4o-mini, Perplexity (pplx-7b-online), and Gemini, then merges values via consensus averages.

Sample Input / Params:

```
{
  "guid": "art-001",
  "title": "Knesset approves new budget",
  "description": "The Israeli parliament approved the 2025 state budget after weeks of negotiations.",
  "source": "Haaretz",
  "source_url": "https://haaretz.com"
}
```

Query: ?user_tier=pro

Sample Output / Response:

```
{
  "guid": "art-001",
  "sentiment": "neutral",
  "bias": "left",
  "bias_score": 0.45,
  "bias_types": ["source leaning", "editorial framing"],
  "bias_category": "Source Bias",
  "bias_score_explanation": "Source Haaretz is registered as left-leaning.",
  "credibility_score": 0.82,
  "credibility_label": "likely credible",
  "fact_check_score": 0.74,
  "summary_hebrew": "הקנעסט איבערנומען דעם 2025.",
  "topics": ["politics", "economy"],
  "claims": ["Budget approved after weeks of negotiations"],
  "factual_points": ["Knesset plenary voted on the bill"],
  "claim_explanation": "Negotiation timeline is framed as an assertion.",
  "bias_explanation": "Article reflects reporting with moderate left-wing editorial frame."
}
```

POST `/ai/analyze/batch` — Batch Analyze Articles

Backend Logic: Accepts a list of up to 20 articles and executes `analyze_article` concurrently using `asyncio.gather`.

Sample Input / Params:

```
{
  "articles": [
    {
      "guid": "art-001",
      "title": "Title 1",
      "description": "Description 1"
    }
  ]
}
```

Query: `?user_tier=free`

Sample Output / Response:

```
[
  {
    "guid": "art-001",
    "sentiment": "neutral",
    "bias": "unknown",
    "bias_score": 0.05,
    "bias_types": ["neutral framing"],
    "bias_category": "Objective Reporting",
    "bias_score_explanation": "Baseline score",
    "credibility_score": 0.5,
    "credibility_label": "needs review",
    "fact_check_score": 0.45,
    "summary_hebrew": "[██████ ████████] Title 1",
    "topics": ["general"],
    "claims": ["Title 1"],
    "factual_points": ["Title 1"],
    "claim_explanation": "Baseline",
    "bias_explanation": "No lean detected"
  }
]
```

GET `/ai/source-bias` — Get Single Source Seed Bias

Backend Logic: Returns the default baseline bias alignment and credibility score for a source from the pre-seeded local lookup table.

Sample Input / Params: `?source=Haaretz&source_url=https://haaretz.com`

Sample Output / Response:

```
{
  "source_name": "Haaretz",
  "bias": "left",
  "credibility_score": 0.82,
  "bias_label": "██ Left"
}
```

GET /ai/source-bias/all — Get Full Baseline Bias Table

Backend Logic: Compiles all preconfigured news sources in ISRAELI_SOURCES, maps their seed bias and credibility tables, and yields a complete sorted dataset.

Sample Input / Params: None

Sample Output / Response:

```
{
  "total": 13,
  "sources": [
    {
      "source_name": "Arutz Sheva",
      "bias": "right",
      "credibility_score": 0.65,
      "bias_label": "➡️ Right"
    }
  ]
}
```

POST /ai/articles/{article_id}/vote/bias — Submit Crowdsourced Article Bias Vote

Backend Logic: Records a user's assessment of an article's bias ('left', 'center', 'right', 'unclear') with their confidence score in the MongoDB 'bias_votes' collection. Real-time consensus computes weight bias using weights: free=1.0, pro=1.5, platinum=2.0 once 5+ user votes are registered; otherwise trusts the AI analysis.

Sample Input / Params:

Path: /ai/articles/art-123/vote/bias

Query: ?user_id=usr-555&user_tier=pro

Body:

```
{
  "bias_assessment": "left",
  "confidence": 0.85,
  "user_notes": "Highly emotive wording regarding budget deficits."
}
```

Sample Output / Response:

```
{
  "status": "recorded",
  "vote_id": "64f1a2b3c4d5e6f7a8b9c0d1",
  "article_id": "art-123",
  "recorded_at": "2026-06-06T09:00:00Z"
}
```

POST `/ai/sources/{source_name}/vote/credibility` — Vote on News Source Credibility

Backend Logic: Submits a rating ('very_low', 'low', 'medium', 'high', 'very_high') mapping to credibility scores (0.1 to 0.95). Stores vote in MongoDB 'credibility_votes' collection. Updates user consensus credibility score on 5+ user submissions.

Sample Input / Params:

Path: `/ai/sources/Haaretz/vote/credibility`
Query: `?user_id=usr-555&user_tier=pro`
Body:

```
{
  "credibility_level": "high",
  "evidence": "Consistent verification processes and standard reporting practices."
}
```

Sample Output / Response:

```
{
  "status": "recorded",
  "vote_id": "64f1a2b3c4d5e6f7a8b9c0d2",
  "source_name": "Haaretz",
  "recorded_at": "2026-06-06T09:01:00Z"
}
```

POST `/ai/articles/{article_id}/flag` — Flag Article for Moderation

Backend Logic: Records a moderation warning flag ('misinformation', 'propaganda', 'biased', 'unreliable_source') in MongoDB 'article_flags' with status 'pending' for review by moderators.

Sample Input / Params:

Path: `/ai/articles/art-123/flag`
Query: `?user_id=usr-555`
Body:

```
{
  "reason": "misinformation",
  "details": "The GDP figures quoted clash directly with official Central Bureau of Statistics bulletins."
}
```

Sample Output / Response:

```
{
  "status": "flagged",
  "flag_id": "64f1a2b3c4d5e6f7a8b9c0d3",
  "article_id": "art-123",
  "reason": "misinformation",
  "message": "Thank you for helping us maintain quality. Our team will review this."
}
```

GET `/ai/articles/{article_id}/votes` — Get Article Voting Stats & Flags

Backend Logic: Queries database bias votes and flags for the requested article, compiling averages and grouping flags by reason (user IDs are masked for privacy).

Sample Input / Params: Path: `/ai/articles/art-123/votes`

Sample Output / Response:

```
{
  "article_id": "art-123",
  "bias_votes_count": 2,
  "flags_count": 1,
  "votes": {
    "bias_votes": [
      { "bias_assessment": "left", "confidence": 0.85, "helpful_count": 0 }
    ],
    "flag_count": 1,
    "flags_by_reason": { "misinformation": 1 }
  }
}
```

POST `/ai/votes/{vote_id}/helpful` — Mark Vote as Helpful (Upvote)

Backend Logic: Increments `helpful_count` of a specific bias or credibility vote by 1 in MongoDB. Surfaces high-quality community feedback.

Sample Input / Params: Path: `/ai/votes/64f1a2b3c4d5e6f7a8b9c0d1/helpful`
Query: `?vote_type=bias`

Sample Output / Response:

```
{
  "status": "upvoted",
  "vote_id": "64f1a2b3c4d5e6f7a8b9c0d1"
}
```

GET /ai/user/{user_id}/vote-history — Get User Voting History & Reputation

Backend Logic: Queries bias_votes, credibility_votes, and article_flags collections to return total counts and the last 5 records of user submissions.

Sample Input / Params: Path: /ai/user/usr-555/vote-history?limit=10

Sample Output / Response:

```
{
  "user_id": "usr-555",
  "reputation": {
    "bias_votes": 12,
    "credibility_votes": 4,
    "flags_submitted": 2
  },
  "recent_activity": {
    "bias_votes": [],
    "credibility_votes": [],
    "flags": []
  }
}
```

GET /ai/votes/leaderboard — Voter Leaderboard

Backend Logic: Aggregates votes in MongoDB grouped by user_id. Summarizes total votes and total helpful upvotes received, sorting top users descending.

Sample Input / Params: ?limit=10

Sample Output / Response:

```
{
  "leaderboard": [
    {
      "_id": "usr-999",
      "total_votes": 150,
      "helpful_votes": 340
    }
  ],
  "description": "Top voters by helpful votes received"
}
```

4. Political Intelligence

POST /political/sync — Sync Knesset API Core Objects

Backend Logic: Fetches and upserts active 25th Knesset members (KNS_PersonToPosition), political factions (KNS_Faction), and committees (KNS_Committee) directly from official Knesset OData endpoints into MongoDB. Computes standard wing leanings (left/center/right).

Sample Input / Params: None

Sample Output / Response:

```
{
  "synced": {
    "mps": 120,
    "parties": 10,
    "committees": 18
  }
}
```

GET /political/mps — List Members of Knesset (MKs)

Backend Logic: Queries active Members of Knesset from MongoDB 'mps' collection. Allows filtering by political party faction name.

Sample Input / Params: ?party_name=Likud

Sample Output / Response:

```
{
  "total": 32,
  "mps": [
    {
      "knesset_id": 200,
      "name": "Benjamin Netanyahu",
      "name_hebrew": "בֶּנְיָמִין נֶטַנְיָהוּ",
      "role": "Prime Minister",
      "is_active": true
    }
  ]
}
```

GET /political/mps/{mp_id} — Get MP Profile with Quotes & Actions

Backend Logic: Fetches MP core profile by Object ID, then queries the database to join up to 50 quote statements (mp_quotes) and 50 logged votes or actions (mp_actions).

Sample Input / Params: Path: /political/mps/64f1a2b3c4d5e6f7a8b9c0d1

Sample Output / Response:

```
{
  "id": "64f1a2b3c4d5e6f7a8b9c0d1",
  "name": "Benjamin Netanyahu",
  "role": "Prime Minister",
  "consistency_score": 0.85,
  "quotes": [],
  "actions": []
}
```

POST /political/mps/{mp_id}/quotes — Add Statement / Quote for MP

Backend Logic: Inserts an MP quote record in 'mp_quotes' collection containing the statement text, date, source URL, and category topic.

Sample Input / Params:

Path: /political/mps/64f1a2b3c4d5e6f7a8b9c0d1/quotes

Body:

```
{
  "quote": "We will pass this economic reform package immediately.",
  "context": "Press conference in Tel Aviv",
  "source_url": "https://www.ynetnews.com/art-econ",
  "topic": "economy",
  "date": "2026-06-01"
}
```

Sample Output / Response:

```
{
  "mp_id": "64f1a2b3c4d5e6f7a8b9c0d1",
  "quote": "We will pass this economic reform package immediately.",
  "topic": "economy",
  "id": "64f1a2b3c4d5e6f7a8b9c0d9"
}
```

POST /political/mps/{mp_id}/actions — Add Action / Vote Record for MP

Backend Logic: Records a legislative vote, speech, or bill initiation in MongoDB 'mp_actions' matching the MP.

Sample Input / Params:

Path: /political/mps/64f1a2b3c4d5e6f7a8b9c0d1/actions

Body:

```
{
  "action": "Voted YES on Economic Deregulation Bill",
  "action_type": "vote",
  "topic": "economy",
  "source_url": "https://knesset.gov.il/vote-101",
  "date": "2026-06-05"
}
```

Sample Output / Response:

```
{
  "mp_id": "64f1a2b3c4d5e6f7a8b9c0d1",
  "action": "Voted YES on Economic Deregulation Bill",
  "topic": "economy",
  "id": "64f1a2b3c4d5e6f7a8b9c0d0"
}
```

POST /political/mps/{mp_id}/contradictions/scan — Scan MP for Statement Contradictions

Backend Logic: Compares all stored quotes and actions for an MP matching identical topics. Triggers GPT-4o-mini to search logical contradictions (or runs rule-based keyword opposition checks as fallback). Records findings in MongoDB and re-calculates MP's reliability 'consistency_score'.

Sample Input / Params: Path: /political/mps/64f1a2b3c4d5e6f7a8b9c0d1/contradictions/scan

Sample Output / Response:

```
{
  "mp_id": "64f1a2b3c4d5e6f7a8b9c0d1",
  "new_contradictions_found": 0,
  "contradictions": []
}
```

GET /political/mps/{mp_id}/contradictions — Get Recorded MP Contradictions

Backend Logic: Retrieves quote-vs-action contradiction documents stored for the requested MP, sorted by detection date.

Sample Input / Params: Path: /political/mps/64f1a2b3c4d5e6f7a8b9c0d1/contradictions

Sample Output / Response:

```
{
  "mp_id": "64f1a2b3c4d5e6f7a8b9c0d1",
  "total": 1,
  "contradictions": [
    {
      "quote_id": "64f1a2b3...",
      "action_id": "64f1a2b4...",
      "explanation": "MP voted against the economic package after declaring support in public.",
      "severity": "high",
      "topic": "economy",
      "detected_at": "2026-06-06T09:10:00Z"
    }
  ]
}
```

GET /political/parties — List Parties & Factions

Backend Logic: Returns all registered political parties from MongoDB 'parties' collection sorted by Knesset seat count.

Sample Input / Params: None

Sample Output / Response:

```
{
  "total": 3,
  "parties": [
    { "name": "Likud", "seats": 32, "wing": "right" },
    { "name": "Yesh Atid", "seats": 24, "wing": "center" },
    { "name": "Labor", "seats": 4, "wing": "left" }
  ]
}
```

GET /political/parties/{party_id} — Get Party Details & Members List

Backend Logic: Retrieves party document metadata and lists active Knesset members matching this party ID from the database.

Sample Input / Params: Path: /political/parties/64f1a2b3c4d5e6f7a8b9c0d2

Sample Output / Response:

```
{
  "id": "64f1a2b3c4d5e6f7a8b9c0d2",
  "name": "Likud",
  "seats": 32,
  "members": [
    { "name": "Benjamin Netanyahu", "role": "Prime Minister" }
  ]
}
```

GET /political/committees — List Knesset Committees

Backend Logic: Queries Knesset standing committees stored in MongoDB, sorting alphabetically.

Sample Input / Params: None

Sample Output / Response:

```
{
  "total": 2,
  "committees": [
    { "committee_id": 1, "name": "Finance Committee", "name_hebrew": "מְדִינַת הַמָּוֶל" }
  ]
}
```

POST /political/bills/sync — Sync Recent Knesset Bills

Backend Logic: Fetches recent bills from Knesset API, parses core fields (status, initiators, committee, summaries), and upserts them into MongoDB 'knesset_bills'.

Sample Input / Params: ?limit=50

Sample Output / Response:

```
{
  "synced": 50
}
```

GET /political/bills/health — Check Official Knesset API Health

Backend Logic: Tests direct connectivity and response latency for the official Knesset OData endpoint, providing a quick debugging ping.

Sample Input / Params: None

Sample Output / Response:

```
{
  "official_api_reachable": true,
  "notice": "Knesset API reachable"
}
```

POST /political/bills/{bill_id}/sync-votes — Sync Official Votes on a Bill

Backend Logic: Queries the official Knesset vote OData collections matching this bill. Iterates through records, resolves Knesset person ID to internal MP profiles, and inserts votes in MongoDB 'bill_vote_records'.

Sample Input / Params: Path: /political/bills/20123/sync-votes

Sample Output / Response:

```
{
  "bill_id": "20123",
  "mp_votes_synced": 120
}
```

GET /political/bills — List Synced Knesset Bills

Backend Logic: Queries local 'knesset_bills' sorted by last updated date, returning summaries. Displays reachability notice of Knesset API.

Sample Input / Params: ?limit=20

Sample Output / Response:

```
{
  "total": 1,
  "bills": [
    { "bill_id": "20123", "name": "State Budget Law", "status": "Approved" }
  ],
  "official_api_reachable": true,
  "official_api_notice": "Connected to official Knesset OData API"
}
```

GET /political/bills/{bill_id} — Get Synced Bill Details & Summaries

Backend Logic: Resolves bill by ID. If summary exists but no AI summary is generated, invokes GPT-4o-mini to compose a summary in Hebrew. Joins user community votes tally and official MK vote tallies.

Sample Input / Params: Path: /political/bills/20123

Sample Output / Response:

```
{
  "bill_id": "20123",
  "name": "State Budget Law",
  "summary": "This bill defines state revenues...",
  "ai_summary": "■■■■ ■■■■■■ ■■■■ 2026...",
  "community_tally": { "total_votes": 10, "support": 8, "oppose": 2, "support_pct": 80.0 },
  "official_vote_summary": { "total": 120, "yes": 64, "no": 56, "abstain": 0, "absent": 0, "other": 0 }
}
```

GET /political/bills/{bill_id}/votes — Get Synced MK Vote Records on Bill

Backend Logic: Retrieves official vote records associated with a bill from 'bill_vote_records', returning vote values (yes/no/abstain) for each MK.

Sample Input / Params: Path: /political/bills/20123/votes

Sample Output / Response:

```
[
  {
    "bill_id": "20123",
    "mp_name": "Benjamin Netanyahu",
    "party": "Likud",
    "vote": "yes",
    "vote_date": "2026-06-05T18:00:00"
  }
]
```

GET /political/mps/{mp_id}/votes — Get Official Voting History for MK

Backend Logic: Queries 'bill_vote_records' matching the MP object ID or Knesset person ID. Returns chronological voting history.

Sample Input / Params: Path: /political/mps/64f1a2b3c4d5e6f7a8b9c0d1/votes

Sample Output / Response:

```
[
  {
    "bill_id": "20123",
    "mp_name": "Benjamin Netanyahu",
    "vote": "yes",
    "vote_date": "2026-06-05T18:00:00"
  }
]
```

POST /political/bills/{bill_id}/vote — Submit Referendum Vote on Bill

Backend Logic: Allows system users to vote on a bill (referendum model). Saves vote in MongoDB 'bill_votes' collection and returns the updated community tally.

Sample Input / Params:

Path:	/political/bills/20123/vote
Body:	<pre>{ "support": true }</pre>

Sample Output / Response:

```
{
  "bill_id": "20123",
  "total_votes": 15,
  "support": 12,
  "oppose": 3,
  "support_pct": 80.0
}
```

GET /political/bills/{bill_id}/tally — Get Community Vote Tally

Backend Logic: Counts support and oppose votes submitted by community members for a bill.

Sample Input / Params:

Path:	/political/bills/20123/tally
-------	------------------------------

Sample Output / Response:

```
{
  "bill_id": "20123",
  "total_votes": 15,
  "support": 12,
  "oppose": 3,
  "support_pct": 80.0
}
```

5. Smart Correlation Engine

POST /correlation/cluster — Cluster Articles by Theme

Backend Logic: Groups input articles by theme and exposes contradictions. - free: Heuristic Jaccard overlap of extracted TF-IDF keywords. Flags contradictions if sources use opposing sentiment keywords (e.g. success vs crisis). - pro: Sends article titles/sources to GPT-4o-mini, yielding semantic clusters and inconsistencies via structured JSON.

Sample Input / Params:

```
{
  "articles": [
    {
      "guid": "a1",
      "title": "Bank of Israel hikes interest rates to combat inflation.",
      "source": "Globes"
    },
    {
      "guid": "a2",
      "title": "Shekel strengthens following Bol interest rate hike.",
      "source": "Haaretz"
    }
  ],
  "use_ai": false
}
```

Sample Output / Response:

```
{
  "total_articles": 2,
  "total_clusters": 1,
  "clusters_with_inconsistency": 0,
  "clusters": [
    {
      "theme": "rates, shekel, interest",
      "article_count": 2,
      "articles": [
        { "guid": "a1", "title": "Bank of Israel..." }
        { "guid": "a2", "title": "Shekel strengthens..." }
      ],
      "inconsistency": null,
      "method": "rule-based"
    }
  ]
}
```

GET /correlation/cluster/live — Cluster Live News Feeds

Backend Logic: Fetches live news from all RSS categories concurrently, flattens articles, clusters them dynamically via keywords or AI, and sorts clusters prioritizing conflicts.

Sample Input / Params: `?limit=15&use_ai=true`

Sample Output / Response:

```
{
  "total_articles": 45,
  "total_clusters": 12,
  "clusters_with_inconsistency": 1,
  "clusters": []
}
```

POST /correlation/cross-reference — Match News to Knesset Bills

Backend Logic: Cross-references input articles with input bills. Analyzes keyword overlap Jaccard scores. Returns matches with `relevance_score` values.

Sample Input / Params:

```
{
  "articles": [
    { "title": "New education reform passes first reading" }
  ],
  "bills": [
    { "name": "Education Reform Act 2025", "name_hebrew": "תקנות חוק חינוך 2025" }
  ]
}
```

Sample Output / Response:

```
{
  "total_articles": 1,
  "articles_with_bill_matches": 1,
  "matches": [
    {
      "article_title": "New education reform passes first reading",
      "related_bills": [
        {
          "name": "Education Reform Act 2025",
          "relevance_score": 0.25
        }
      ]
    }
  ]
}
```

GET /correlation/cross-reference/live — Cross-reference Live News & Bills

Backend Logic: Fetches live news articles and Knesset bills concurrently. Compares and indexes matches automatically.

Sample Input / Params: ?limit=15

Sample Output / Response:

```
{
  "total_articles": 30,
  "total_bills": 20,
  "articles_with_bill_matches": 2,
  "matches": []
}
```

6. AI Insights & Trends

GET /insights/trends — Detect Global Trending Topics

Backend Logic: Aggregates titles from live news. Filters common stop words. Computes keyword frequencies. If use_ai=true, triggers GPT-4o-mini for semantic trends synthesis.

Sample Input / Params: ?limit=10&use_ai=false

Sample Output / Response:

```
{
  "total_articles_analyzed": 120,
  "use_ai": false,
  "trends": [
    { "keyword": "budget", "count": 14, "sample_headlines": [] }
  ]
}
```

GET /insights/all — Generate Insights for All Categories

Backend Logic: Concurrently fetches news and generates category insights. Compiles global trends from the combined keywords list.

Sample Input / Params: ?limit=10&use_ai=true

Sample Output / Response:

```
{
  "economy": {
    "category": "economy",
    "insight_hebrew": "■■■■ ■■■■■ ■■■■ ■■■■ ■■■■..."
  },
  "global_trends": [
    { "keyword": "budget", "total_count": 25 }
  ]
}
```

GET /insights/{category} — Generate Category Insights

Backend Logic: Extracts trending keywords for category articles. Generates a 3-sentence summary insight in Hebrew (using GPT-4o-mini if use_ai=true, or rule-based English descriptors as fallback). Responses are cached.

Sample Input / Params: Path: /insights/economy
Query: ?limit=20&use_ai=true

Sample Output / Response:

```
{
  "category": "economy",
  "article_count": 20,
  "insight_hebrew": "התחזית הכלכלית מראה המשך צמיחה מתונה, אך עם סיכונים מוגדלים בשל גורמים חיצוניים. הממשלה מתכננת להתמודד עם אתגרים אלו באמצעות מדיניות מאוזנת.",
  "trends": [
    { "keyword": "inflation", "count": 6 }
  ],
  "generated_at": "2026-06-06T09:12:00Z"
}
```

GET /insights/subscription/summary/{period} — Generate Subscription Digest

Backend Logic: Assembles a lightweight text digest of the top three headlines from each news category. Period must be 'weekly' or 'monthly'. Designed for newsletter dispatch.

Sample Input / Params: Path: /insights/subscription/summary/weekly?limit=5

Sample Output / Response:

```
{
  "period": "weekly",
  "generated_at": "2026-06-06T09:12:00Z",
  "total_headlines": 18,
  "summary": "Economy: BoI rates steady | Tech investment rises\nPolitics: Knesset budget debate..."
}
```

7. AI Q&A; Box

POST /qa/ask — Ask GPT-4o with News Context

Backend Logic: Accepts a natural language query. Formats context using recent articles matching the requested category. Configures a system prompt directing GPT-4o to act as a factual, neutral Israeli analyst. Caches responses.

Sample Input / Params:

```
{
  "question": "What is the status of the education budget?",
  "category": "education",
  "use_context": true,
  "model": "gpt-4o"
}
```

Sample Output / Response:

```
{
  "answer": "According to Ynet and Kan news, the education budget is projected to increase by 4% to fund school infrastructure...",
  "model": "gpt-4o",
  "tokens_used": 412,
  "cached": false
}
```

POST /qa/ask/stream — Stream Q&A; Responses (SSE)

Backend Logic: Accepts Q&A; body, constructs system contexts, and establishes a Server-Sent Events (SSE) stream (text/event-stream) streaming token increments.

Sample Input / Params:

```
{
  "question": "Explain Knesset budget updates",
  "model": "gpt-4o-mini"
}
```

Sample Output / Response:

```
data: According<br/>data: to recent<br/>data: updates...<br/>data: [DONE]
```

POST /qa/ask/ithy — Ask ITHY Simulated Multi-Perspective AI

Backend Logic: Simulates the ITHY.ai web-search multi-perspective engine. Invokes GPT-4o with instructions to search angles, detail conflicting opinions, and build syntheses.

Sample Input / Params:

```
{
  "question": "What are the arguments surrounding the shekel stability?"
}
```

Sample Output / Response:

```
{
  "answer": "Analysis from three key perspectives:\n1. Exporters claim shekel strength hurts competitiveness...\n2. Central bank states shekel strength curbs imported inflation...\n3. Retailers cite benefits of cheaper import pricing...",
  "source": "ithy-simulated",
  "cached": false
}
```

POST /qa/ask/dual — Compare GPT vs ITHY Side-by-Side

Backend Logic: Triggers ask_gpt and ask_ithy concurrently. Combines answers, and prompts gpt-4o-mini with a summarization prompt to produce a one-sentence comparative differential note.

Sample Input / Params:

```
{
  "question": "Is the government coalition stable?",
  "model": "gpt-4o"
}
```

Sample Output / Response:

```
{
  "question": "Is the government coalition stable?",
  "gpt": {
    "answer": "The coalition holds a majority but faces policy tension...",
    "model": "gpt-4o"
  },
  "ithy": {
    "answer": "From political analysts, stability is viewed through party cohesion..."
  },
  "comparison_note": "GPT highlights specific policy issues while ITHY emphasizes general party cohesion."
}
```

GET /qa/ask — Quick GET Q&A; Endpoint

Backend Logic: GET query endpoint wrapper. Resolves recent database articles as context, then calls gpt-4o-mini.

Sample Input / Params: ?q=What+happened+in+Knesset+today

Sample Output / Response:

```
{
  "answer": "Today, the Knesset debated educational funding bills...",
  "model": "gpt-4o-mini",
  "tokens_used": 240,
  "cached": false
}
```

8. Social Media Integration

GET /social/twitter/search — Search Twitter / X via twscrape

Backend Logic: Invokes twscrape query parser. If `israeli_only=true`, builds query appending preconfigured Israeli accounts and hashtags. Maps responses to NewsArticle schema. Returns 503 if twscrape is not installed.

Sample Input / Params: ?query=Israel+economy&limit=10&israeli_only=true

Sample Output / Response:

```
{
  "query": "Israel economy",
  "full_query": "(Israel economy) (from:Jerusalem_Post OR from:TimesofIsrael) lang:en",
  "count": 1,
  "tweets": [
    {
      "title": "Tweet by @Jerusalem_Post",
      "link": "https://twitter.com/Jerusalem_Post/status/12999",
      "description": "Israeli economy expands 3% in Q1 according to official statistics.",
      "pub_date": "2026-06-06T08:00:00Z",
      "source": "Twitter / X",
      "source_url": "https://twitter.com/Jerusalem_Post",
      "guid": "12999"
    }
  ]
}
```

GET /social/twitter/user/{username} — Get User Tweets via twscrape

Backend Logic: Resolves Twitter username login ID to numeric Twitter user ID using twscrape. Fetches user's recent tweets.

Sample Input / Params: Path: /social/twitter/user/TimesofIsrael?limit=5

Sample Output / Response:

```
{
  "username": "TimesofIsrael",
  "count": 1,
  "tweets": []
}
```

GET /social/twitter/israeli-accounts — Get Israeli Account Tweets

Backend Logic: Iterates parallel calls via asyncio.gather fetching recent tweets from all 10 preconfigured active accounts.

Sample Input / Params: ?limit=5

Sample Output / Response:

```
{
  "accounts": ["Jerusalem_Post", "TimesofIsrael", "haaretzcom", ...],
  "data": {
    "TimesofIsrael": []
  }
}
```